INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

💻 Infosys DSE & Specialist Programmer (SP) — Architecture, Syllabus & Exam Blueprint

Section: 07_[OPTIONAL_DSE_SP_ONLY]_Coding_Prep | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

💻 Infosys DSE & Specialist Programmer (SP) — Architecture, Syllabus & Exam Blueprint

> **Role Streams:**
> 1. **Systems Engineer (SE):** 3.6 – 4.0 LPA (Aptitude + Pseudocode + Puzzles + Verbal + Essay)
> 2. **Digital Specialist Engineer (DSE):** 6.5 LPA (Advanced Hands-on Coding + Medium DSA)
> 3. **Specialist Programmer (SP / Power Programmer):** 9.5 LPA (Hard Competitive Programming, Advanced Graphs, DP)
>
> **Primary Selection Channels:** HackWithInfy, InfyTQ / Springboard Certification Upgrade, On-Campus DSE/SP Track, Internal SE-to-DSE Bridge Exam

1. DSE & SP Examination Pattern

Assessment Track	Questions	Time Limit	Languages Supported	Problem Difficulty	Passing Expectation
DSE Coding Round	**3 Problems**	**180 Minutes**	Java, Python, C++, C#	1 Easy + 1 Medium + 1 Upper Medium	Solve 2 to 2.5 Problems Completely
SP / Power Programmer	**3 Problems**	**180 Minutes**	Java, Python, C++, C#	1 Medium + 2 Hard (DP / Graph / Trees)	Solve 2+ Problems Completely with Optimal Time Complexity

Evaluation Environment & Platform:

- Strict sandbox execution with hidden test cases evaluating edge cases (negative values, large $N \leq 10^5$, memory constraints).
- Strict $O(N \log N)$ or $O(N)$ time complexity limits preventing brute force $O(N^2)$ solutions from passing full test suites.

2. Core Algorithmic Domains & Syllabus

Domain 1: Arrays, Strings, Two-Pointers & Sliding Window

- Kadane's Algorithm (Maximum Subarray Sum with variations).
- Longest Substring Without Repeating Characters / Subarray with given sum.
- Trapping Rain Water, Container With Most Water, 3Sum and 4Sum variants.
- Prefix sums, Difference arrays, and 2D matrix rotations.

Domain 2: Dynamic Programming (DP) & Memoization

- 0/1 Knapsack Problem, Unbounded Knapsack, Fractional Knapsack.
- Longest Increasing Subsequence (LIS in $O(N \log N)$) & Longest Common Subsequence (LCS).
- Matrix Chain Multiplication (MCM) & Partition Equal Subset Sum.
- Coin Change (Min coins and Total ways), Word Break, House Robber.

Domain 3: Trees, Binary Search Trees & Graphs

- Lowest Common Ancestor (LCA) in Binary Trees and BSTs.
- Level order spiral traversal, Vertical order traversal, Diameter of a Tree.
- Breadth-First Search (BFS) & Depth-First Search (DFS) on grid mazes and component graphs.
- Dijkstra's Shortest Path Algorithm, Bellman-Ford, Floyd-Warshall.
- Topological Sort (Kahn's Algorithm) and Cycle Detection in Directed/Undirected Graphs.
- Disjoint Set Union (DSU / Kruskal's Algorithm) for Minimum Spanning Trees.

Domain 4: Backtracking & Greedy Algorithms

- N-Queens Problem, Rat in a Maze, Sudoku Solver.
- Word Search in 2D Board, Subset Sum / Permutations generation.
- Activity Selection, Job Sequencing with Deadlines, Huffman Coding.